

BioSpecInfo

Data Model — BioSpecInfo

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Data Model — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	bsi-v197
Purpose	Document where the data live, with what structure, and what lifecycle they have.

0. A necessary premise

BioSpecInfo has no server database. There is no backend, so there are no centrally managed relational tables, no migrations, and no users stored anywhere.

There is, however, a real data model, and an assessor is entitled to see it documented. It is organised across three levels:

Level	Technology	What it holds	Where it lives
Preferences and progress	localStorage	Settings, API keys, AI history, study progress	The user's browser
Document archive	IndexedDB	Photos, documents, HTML sections, notes, activity log	The user's browser
Teaching datasets	SQLite via sql.js	Exercise tables, created in memory at each session	Memory, not persisted

All data stay on the device. They are not transmitted, there is no server-side user identifier, and the user can delete them selectively from the interface.

1. Entity diagram

```
erDiagram
    DISPOSITIVO ||--|| PREFERENZE : "possiede"
    DISPOSITIVO ||--|| ARCHIVIO : "possiede"
    DISPOSITIVO ||--o{ SESSIONE_SQL : "apre"

    PREFERENZE {
        string bsi_ai_provider "fornitore AI scelto"
        json bsi_api_keys "chiavi per fornitore"
        string bsi_proxy_url "indirizzo del proxy"
        json bsi_ai_threads "conversazioni salvate"
        json bsi_ai_memory "memoria dell-agente"
        json bsi_prov_ko "fornitori irraggiungibili"
        json bsi_modelli_ko "modelli ritirati"
        json bsi_tetti "tetti di token appresi"
        json bsi_srs "ripetizione spaziata"
        json bsi_quiz_progress "avanzamento quiz"
        string bsi_section "ultima sezione aperta"
    }

    ARCHIVIO ||--o{ PHOTOS : contiene
    ARCHIVIO ||--o{ DOCS : contiene
    ARCHIVIO ||--o{ SECTIONS : contiene
    ARCHIVIO ||--o{ NOTES : contiene
    ARCHIVIO ||--o{ ACTIVITY : registra

    PHOTOS {
        int id PK "autoIncrement"
        string name
        string type "MIME"
        int size "byte"
        string data "data URL"
        int added "epoch ms"
    }
    DOCS {
        int id PK "autoIncrement"
        string name
        string type
        int size
        string data "data URL"
        int date "epoch ms"
    }
    SECTIONS {
        int id PK "autoIncrement"
        string name
        string icon
        string html "contenuto"
        int size "caratteri"
        int added "epoch ms"
    }
    NOTES {
        string key PK "sempre main"
        string text
        int updated "epoch ms"
    }
    ACTIVITY {
        int id PK "autoIncrement"
        string icon
```

```

string msg
string time "data locale"
}

```

The diagram keeps the entity and field names as they appear in the code, in Italian: renaming them in the translation would document a model that does not exist.

2. Preferences and state — localStorage

2.1 AI assistant

Key	Type	Content	Lifecycle
bsi_ai_provider	string	Identifier of the selected provider	Until the user changes it
bsi_api_keys	JSON	{ providerId: key }	Until explicitly deleted
bsi_api_key	string	Single key, legacy format	Migrated to bsi_api_keys
bsi_proxy_url	string	Proxy address	Until "Disconnect"
bsi_ai_threads	JSON	{ threads: [...], activeId } — max 30 chats, 100 messages each	Pruned automatically
bsi_ai_memory	JSON	Facts the agent remembers between sessions	Deletable
bsi_ai_history	JSON	Condensed history	Deletable
bsi_nucleo	string	"Core" mode active	—

2.2 What the assistant learns about providers

These three keys are the reason the assistant improves with use instead of repeating the same mistakes. They have an expiry because today's failure must not be a permanent sentence.

Key	Structure	Expiry	What it is for
bsi_prov_ko	{ providerId: { t: epoch, n: count } }	24 hours	Providers that do not answer from this device. Demoted in the selection, never removed.
bsi_modelli_ko	{ digest: { modelli: [...], t } }	7 days	Models retired by the provider, so as not to offer them again. Indexed on the digest of the key.
bsi_tetti	{ providerId: tokenCeiling }	Permanent	Token ceiling learned from the error message rather than hard-coded

2.3 Study and progress

Key	Content
bsi_srs	Spaced repetition: cards, intervals, next review

Key	Content
<code>bsi_sm2</code>	Parameters of the SM-2 algorithm
<code>bsi_quiz_progress</code> , <code>bsi_quiz_history</code>	Quiz progress and history
<code>bsi_studypath</code>	The study path chosen
<code>bsi_notes</code> , <code>bsi_note_<section></code>	Personal notes, global and per section
<code>bsi_ds_prog</code> , <code>bsi_ds_proj</code>	Progress in the data-science module
<code>bsi_section</code>	Last section opened, to resume from there
<code>bsi_guide_v3</code>	Introductory guide already seen

2.4 Identifiers and licence

Key	Content	Privacy note
<code>bsi_device_id</code>	Random local identifier	Not transmitted. It only distinguishes installations within the same browser.
<code>bsi_user_email</code>	Email, if entered by the user	Entered voluntarily, not transmitted
<code>bsi_pro_license</code> , <code>bsi_trial_start</code>	State of Pro mode	Local
<code>bsi_rdkit_lib2</code>	Molecule library of the laboratory	Local

That none of these ever leaves the device is verified directly by `audit_rete` , which seeds a canary value into these very keys and then inspects every outgoing request.

3. Document archive — IndexedDB

Database name	<code>bsi_filemanager_v1</code>
Schema version	<code>2</code>
Object stores	<code>photos</code> , <code>docs</code> , <code>sections</code> , <code>notes</code> , <code>activity</code>

3.1 Structure of the stores

Store	Key	Fields	Notes
<code>photos</code>	<code>id</code> (autoIncrement)	<code>name</code> , <code>type</code> , <code>size</code> , <code>data</code> , <code>added</code>	Images as data URLs. Per-file limit: 20 MB
<code>docs</code>	<code>id</code> (autoIncrement)	<code>name</code> , <code>type</code> , <code>size</code> , <code>data</code> , <code>date</code>	Generic documents
<code>sections</code>	<code>id</code> (autoIncrement)	<code>name</code> , <code>icon</code> , <code>html</code> , <code>size</code> , <code>added</code>	HTML pages uploaded by the user

Store	Key	Fields	Notes
notes	key	key ("main"), text , updated	A single entry
activity	id (autoIncrement)	icon , msg , time	Log of the most recent operations

The `files` store, present in earlier schema versions, is no longer created from version 2.

3.2 Opening and unavailability

Opening IndexedDB is **asynchronous**, and in some configurations (Safari in private browsing, site data blocked) it **never happens**. The code awaits a promise rather than reading a variable that might not yet be set, and distinguishes two cases:

- **reading** with the store absent → returns an empty set, the page shows

"nothing yet";

- **writing** with the store absent → the error reaches whoever pressed the

button, and a persistent warning appears at the top of the page saying nothing will be kept.

If another tab requests a version upgrade, the connection is closed (`onversionchange`) so as not to block both.

4. Teaching datasets — SQLite in memory

The data-exercise module creates an SQLite database **in memory** through `sql.js` (SQLite compiled to WebAssembly). The tables are generated at each session from the datasets bundled in the repository and are **not persisted**: closing the tab makes them vanish.

The choice is deliberate: the exercise must start from a known state every time, and no user data end up in a database they did not create.

5. Reference data — constants in the code

The scientific data are not in a database but in constant structures inside `index.html` , loaded with the application. They are read-only and versioned together with the code.

Structure	Content	Verification
FARM_DATA	263 drugs : name, category, SMILES, molecular weight, mechanism of action, indications, adverse effects, class; 36 entries also carry <code>chembl</code> and the record's <code>formula</code>	<code>audit_farmaci</code> — structure ↔ weight, duplicates, and the formula rebuilt from the graph against ChEMBL's
UV_DATA	Chromophores, $\lambda_{\max}$, extinction coefficients, Woodward's rules	<code>audit_dati</code>
MS_DATA	Characteristic fragmentations by class	<code>audit_dati</code>

Structure	Content	Verification
MOLECOLE_ESAME	Tabulated exam molecules	test_spettri
PHYS_SIMBOLI	Physical constants with symbol and unit	test_costanti
SMARTS , SMARTS_H	Patterns for functional-group recognition	test_spettri

6. Lifecycle and deletion

The user can delete data **by group**, from the interface, without wiping everything. The groups are defined in `bsiCancellaDati()`.

Group	What it removes
chiavi	API keys, chosen provider, proxy address, memory of unreachable providers
chat	Conversations and the assistant's memory
studio	Spaced repetition, quizzes, study path, notes
tutto	Every <code>bsi_*</code> key and the entire IndexedDB archive

Deleting the archive proceeds even if a single entry fails, and reports how many were not removed: a "delete everything" operation that stops halfway without saying so would be worse than the problem it solves.

7. Retention and space limits

Constraint	Value	Behaviour when exceeded
localStorage	5–10 MB depending on the browser	Writes fail. The app stays operational: chats are pruned (30 chats, 100 messages), and every write is guarded.
IndexedDB	Quota granted by the browser	Error reported to whoever pressed, with a persistent warning
Single image	20 MB	Refused with an explicit message; the upload of the others continues

Behaviour with space exhausted is verified by the `audit_quota` and `test_filemanager` benches: ten pages out of ten stay usable with `localStorage.setItem` forced to fail.

Document updated to version `bsi-v197`.